Solving the Bellman Equations for a Stochastic Policy in Sutton & Barto Notation

1 The Problem

Consider a finite Markov decision process with state set $\mathcal{S} = \{s_1, s_2, s_3\}$ and action set $\mathcal{A} = \{a, b\}$. We use the discounted return

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1},$$

with discount rate $\gamma = 0.9$.

Stochastic policy. Suppose the agent follows the stochastic policy

	$\pi(a \mid s)$	$\pi(b \mid s)$
s_1	0.5	0.5
s_2	0.5	0.5
s_3	0.5	0.5

so at every state the agent chooses a with probability 0.5 and b with probability 0.5. We want the state-value function $v_\pi(s)$ of this policy,

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]\,,$$

for every $s \in \mathcal{S}$.

2 Transition Dynamics

The dynamics are fully specified by the joint probability

$$p(s', r \mid s, a) = \Pr\{S_{t+1} = s',\, R_{t+1} = r \mid S_t = s,\, A_t = a\}.$$

Table 1 lists every (s, a, s', r) combination with nonzero probability. Combinations not listed have $p(s', r \mid s, a) = 0$. For each (s, a) pair the listed probabilities sum to 1; the reward is deterministic given (s, a), so for each (s, a) all the probability mass over rewards sits at a single value $r(s, a)$.

Equivalently, the expected immediate reward $r(s, a) = \sum_r r \sum_{s'} p(s', r \mid s, a)$ is

$$r(s,a) = \begin{array}{c|cc} & a & b \\ \hline s_1 & 1 & 0 \\ s_2 & 0 & 2 \\ s_3 & 0 & 0 \end{array}$$

Table 1: The dynamics $p(s', r \mid s, a)$. Each row is a possible transition; rows not shown have probability zero.

s	a	s'	r	$p(s', r \mid s, a)$
s_1	a	s_1	1	1
s_1	b	s_1	0	$\frac{1}{2}$
s_1	b	s_2	0	$\frac{1}{2}$
s_2	a	s_1	0	1
s_2	b	s_2	2	1
s_3	a	s_1	0	$\frac{1}{2}$
s_3	a	s_2	0	$\frac{1}{2}$
s_3	b	s_2	0	1

and the marginal transition probabilities $p(s' \mid s, a) = \sum_r p(s', r \mid s, a)$ are

$$P^a = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}, \qquad P^b = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where row i corresponds to s_i and column j to s_j.

Remarks. No transition ever enters s_3 (the third column of both matrices is zero), and the only positive-probability transition out of s_3 under b goes to s_2. Thus s_3 is never entered and plays the role of a terminal state.

3 The Bellman Equation for v_π

The general form

In Sutton & Barto notation, the Bellman equation for the value function of a fixed policy π is

$$v_\pi(s) = \sum_a \pi(a \mid s) \sum_{s', r} p(s', r \mid s, a) \left[r + \gamma \, v_\pi(s') \right], \qquad \forall s \in \mathcal{S}. \tag{1}$$

The outer sum is over actions $a \in \mathcal{A}$; the inner sum is over all next states $s' \in \mathcal{S}$ and rewards r.

Deriving the compact form

We reduce (1) to a form that makes the linear system easy to read.

Step 1: distribute the inner sum. The factor $\left[r + \gamma v_\pi(s') \right]$ splits into two pieces:

$$\sum_{s', r} p(s', r \mid s, a) \left[r + \gamma \, v_\pi(s') \right] = \underbrace{\sum_{s', r} p(s', r \mid s, a) \, r}_{\text{(i)}} \; + \gamma \underbrace{\sum_{s', r} p(s', r \mid s, a) \, v_\pi(s')}_{\text{(ii)}}.$$

Step 2: simplify (i) and (ii). In this MDP the reward is a deterministic function of (s, a) alone: given (s, a), the reward does not depend on which s' is reached. Formally, for each (s, a) there is a unique value $r(s, a)$ such that

$$p(s', r \mid s, a) = \begin{cases} p(s' \mid s, a), & r = r(s, a), \\ 0, & r \neq r(s, a). \end{cases}$$

All the probability mass over rewards sits at the single value $r(s, a)$, so the sum over r collapses.

Term (i). Since the only r with nonzero probability is $r(s, a)$,

$$\sum_{s',r} p(s', r \mid s, a)\, r = \sum_{s'} p(s' \mid s, a)\, r(s, a) = r(s, a) \sum_{s'} p(s' \mid s, a) = r(s, a),$$

using $\sum_{s'} p(s' \mid s, a) = 1$. This is the expected immediate reward.

Term (ii). Similarly, the sum over r picks up only $r(s, a)$ and leaves the marginal $p(s' \mid s, a)$:

$$\sum_{s',r} p(s', r \mid s, a)\, v_\pi(s') = \sum_{s'} p(s' \mid s, a)\, v_\pi(s').$$

Step 3: reassemble. Substituting (i) and (ii) into (1):

$$v_\pi(s) = \sum_a \pi(a \mid s) \Big[r(s, a) + \gamma \sum_{s'} p(s' \mid s, a)\, v_\pi(s') \Big]. \tag{2}$$

The sum over r is gone, and the reward is replaced by its expected value $r(s, a)$.

Why this simplification is legitimate. The key fact is that the reward is a deterministic function of (s, a): once the agent is in state s and picks action a, the reward R_{t+1} is fixed at $r(s, a)$ regardless of which next state is reached. If the reward depended on s' as well, we could not pull r out of the sum over s', and the compact form would be invalid.

Equation (2) is a linear system in the unknowns $v_\pi(s_1), v_\pi(s_2), v_\pi(s_3)$. Because the policy is stochastic, the sum over a does *not* collapse: both actions contribute at every state, each weighted by $\pi(a \mid s)$.

4 Writing Out the Equations

Substituting $\pi(a \mid s) = \pi(b \mid s) = 0.5$ and $\gamma = 0.9$ into (2), using the dynamics in Table 1:

State s_1.

$$\begin{aligned} \text{action } a: &\quad r(s_1, a) + 0.9 \sum_{s'} p(s' | s_1, a)\, v_\pi(s') = 1 + 0.9\, v_\pi(s_1), \\ \text{action } b: &\quad r(s_1, b) + 0.9 \sum_{s'} p(s' | s_1, b)\, v_\pi(s') = 0 + 0.9\big(\tfrac{1}{2}\, v_\pi(s_1) + \tfrac{1}{2}\, v_\pi(s_2)\big). \end{aligned}$$

Averaging with weights $\pi(a|s_1) = \pi(b|s_1) = 0.5$:

$$v_\pi(s_1) = 0.5\Big[1 + 0.9\, v_\pi(s_1)\Big] + 0.5\Big[0 + 0.9\big(\tfrac{1}{2}\, v_\pi(s_1) + \tfrac{1}{2}\, v_\pi(s_2)\big)\Big]. \tag{1}$$

State s_2.

$$\text{action } a: \quad r(s_2,a) + 0.9 \sum_{s'} p(s'|s_2,a)\, v_\pi(s') = 0 + 0.9\, v_\pi(s_1),$$
$$\text{action } b: \quad r(s_2,b) + 0.9 \sum_{s'} p(s'|s_2,b)\, v_\pi(s') = 2 + 0.9\, v_\pi(s_2).$$

Averaging with weights 0.5 and 0.5:

$$v_\pi(s_2) = 0.5\Big[0 + 0.9\, v_\pi(s_1)\Big] + 0.5\Big[2 + 0.9\, v_\pi(s_2)\Big]. \tag{2}$$

State s_3.

$$\text{action } a: \quad r(s_3,a) + 0.9 \sum_{s'} p(s'|s_3,a)\, v_\pi(s') = 0 + 0.9\big(\tfrac{1}{2}\, v_\pi(s_1) + \tfrac{1}{2}\, v_\pi(s_2)\big),$$
$$\text{action } b: \quad r(s_3,b) + 0.9 \sum_{s'} p(s'|s_3,b)\, v_\pi(s') = 0 + 0.9\, v_\pi(s_2).$$

Averaging with weights 0.5 and 0.5:

$$v_\pi(s_3) = 0.5\Big[0 + 0.9\big(\tfrac{1}{2}\, v_\pi(s_1) + \tfrac{1}{2}\, v_\pi(s_2)\big)\Big] + 0.5\Big[0 + 0.9\, v_\pi(s_2)\Big]. \tag{3}$$

5 Simplifying the Equations

Equation (1).

$$v_\pi(s_1) = 0.5 + 0.45\, v_\pi(s_1) + 0.225\, v_\pi(s_1) + 0.225\, v_\pi(s_2).$$

Collecting terms in $v_\pi(s_1)$:

$$0.325\, v_\pi(s_1) - 0.225\, v_\pi(s_2) = 0.5. \tag{1'}$$

Equation (2).

$$v_\pi(s_2) = 0.45\, v_\pi(s_1) + 1 + 0.45\, v_\pi(s_2),$$

so

$$-0.45\, v_\pi(s_1) + 0.55\, v_\pi(s_2) = 1. \tag{2'}$$

Equation (3).

$$v_\pi(s_3) = 0.225\, v_\pi(s_1) + 0.225\, v_\pi(s_2) + 0.45\, v_\pi(s_2) = 0.225\, v_\pi(s_1) + 0.675\, v_\pi(s_2),$$

so

$$v_\pi(s_3) - 0.225\, v_\pi(s_1) - 0.675\, v_\pi(s_2) = 0. \tag{3'}$$

6 Solving the Linear System

Equations (1') and (2') form a 2×2 system in $v_\pi(s_1), v_\pi(s_2)$:

$$0.325\, v_\pi(s_1) - 0.225\, v_\pi(s_2) = 0.5, \tag{1'}$$
$$-0.45\, v_\pi(s_1) + 0.55\, v_\pi(s_2) = 1. \tag{2'}$$

Solve for $v_\pi(s_1)$ and $v_\pi(s_2)$. Multiply (1') by 0.55 and (2') by 0.225:

$$0.17875\, v_\pi(s_1) - 0.12375\, v_\pi(s_2) = 0.275,$$
$$-0.10125\, v_\pi(s_1) + 0.12375\, v_\pi(s_2) = 0.225.$$

Adding eliminates $v_\pi(s_2)$:

$$0.0775\, v_\pi(s_1) = 0.5 \implies v_\pi(s_1) = \frac{0.5}{0.0775} = \frac{1000}{155} = \frac{200}{31} \approx 6.4516.$$

Substituting into (2'):

$$-0.45 \cdot \frac{200}{31} + 0.55\, v_\pi(s_2) = 1 \implies 0.55\, v_\pi(s_2) = 1 + \frac{90}{31} = \frac{121}{31},$$

so

$$v_\pi(s_2) = \frac{121}{31 \cdot 0.55} = \frac{121}{17.05} = \frac{220}{31} \approx 7.0968.$$

Solve for $v_\pi(s_3)$. From (3'):

$$v_\pi(s_3) = 0.225 \cdot \frac{200}{31} + 0.675 \cdot \frac{220}{31} = \frac{45}{31} + \frac{148.5}{31} = \frac{45}{31} + \frac{297}{62} = \frac{387}{62} \approx 6.2419.$$

7 Result

The value function of the stochastic policy $\pi(a \mid s) = \pi(b \mid s) = 0.5$ for all s is

$$\boxed{v_\pi(s_1) = \frac{200}{31} \approx 6.45, \qquad v_\pi(s_2) = \frac{220}{31} \approx 7.10, \qquad v_\pi(s_3) = \frac{387}{62} \approx 6.24.}$$

Check. Substitute back into (1)–(3).

$$\text{Eq. (1): } 0.5\Big[1 + 0.9 \cdot \tfrac{200}{31}\Big] + 0.5\Big[0.9\big(\tfrac{1}{2} \cdot \tfrac{200}{31} + \tfrac{1}{2} \cdot \tfrac{220}{31}\big)\Big] = \tfrac{200}{31}. \quad \checkmark$$

$$\text{Eq. (2): } 0.5\Big[0.9 \cdot \tfrac{200}{31}\Big] + 0.5\Big[2 + 0.9 \cdot \tfrac{220}{31}\Big] = \tfrac{220}{31}. \quad \checkmark$$

$$\text{Eq. (3): } 0.5\Big[0.9\big(\tfrac{1}{2} \cdot \tfrac{200}{31} + \tfrac{1}{2} \cdot \tfrac{220}{31}\big)\Big] + 0.5\Big[0.9 \cdot \tfrac{220}{31}\Big] = \tfrac{387}{62}. \quad \checkmark$$

Interpretation. Because the policy is stochastic, every state mixes the consequences of both actions. The value of s_1 is much lower than under the deterministic b-policy: half the time the agent takes action a, which locks it into the modest reward of 1 per step instead of reaching the more rewarding s_2. Similarly, s_2 is dragged down from 20 (the pure-b value) to about 7.10, because half the time the agent takes action a and is sent back to s_1. The state s_3 is also lower because action a at s_3 scatters the agent between s_1 and s_2 rather than deterministically sending it to s_2. This illustrates how a stochastic policy interpolates between the values of its constituent actions: the Bellman equation is a weighted average over actions rather than a single branch.